Dynamics Modeling and Control of
Multi-Segment Passively-Articulated Autonomous Wheeled Vehicles

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ABSTRACT

The purpose of this thesis is to describe a class of multi-segment, passively-articulated autonomous wheeled vehicles designed for long distance ground transportation across snow surfaces in the polar regions. Uncrewed ground vehicles (UGVs) are currently used in the polar regions for data collection over terrain that would be dangerous for crewed vehicles. The range of snow conditions that may be encountered in the polar regions is vast and hard to fully predict. Of the conditions that can be expected, traditional wheeled mobility is not always sufficient, even with sophisticated methods for immobilization detection and prevention. The obstacles to mobility in the polar regions are unique in that they often give no visual indications of their presence and can only be measured proprioceptively. Therefore, avoiding hazards is less of a focus, and UGVs must instead rely on robust mobility systems that can overcome or prevent imminent immobilizations. The risk of immobilization is one factor preventing greater use of UGVs in long distance applications.

This thesis presents one solution to the polar mobility challenge in the form of the *SnoWorm* UGV. The *SnoWorm* is a high mobility UGV made up of multiple four-wheeled segments that are passively articulated relative to one another. *SnoWorm*'s long length and other features give it superior mobility on snow surfaces when compared to similar wheeled vehicles. *SnoWorm* is simulated within the Gazebo robotics simulator using a custom terrain model that provides net force versus slip characteristics comparable to more sophisticated terramechanics models for snow. Data collected from a physical UGV operating on snow is used to ground-truth the terrain model for both steady state driving and immobilizations in poorly-bonded snow. Multiple control algorithms are explored, one allowing the *SnoWorm* to drive in a linear manner, where segments follow one another, and another allowing lateral "crab" movement that minimizes repeated passes over the same terrain. Measuring the forces between segments gives a better understanding of vehicle dynamics and allows for control schemes to vary the mobility required from individual segments.

While increasing vehicle length and the number of wheels does improve mobility over a variety of snow surfaces, this is insufficient for decreasing the risk of immobilization to the point where long distance uses of UGVs become feasible. Tracks instead of wheels help, but their much lower efficiency compared to wheels is especially relevant for UGVs covering long distances. Therefore, "inchworming", an alternative mode of locomotion for the *SnoWorm*, is evaluated in this thesis. Inchworming relies on powered prismatic joints between segments to enable them to push and pull one another. This inchworm technique vastly increases the total range of conditions over which the vehicle is able to make forward progress, with minimal added complexity.

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Contents

List of Tables

List of Figures

Nomenclature

$\vec{a}$	Acceleration vector	m/s^2
b_{fT}	Coefficient for frictional torque from wheel speed	$N \cdot m \cdot s/rad$
b_{yrM}	Coefficient for yaw resistive moment about axle Z-axis	$N \cdot m \cdot s/rad$
b_{trT}	Coefficient for terrain resistance torque	N/s
b_{tbF}	Coefficient for tractive force versus slip	$unitless$
b_{ltrF}	Coefficient for lateral terrain resistance force	$unitless$
b_{trF}	Coefficient for longitudinal terrain resistance force	$unitless$
F_n	Normal force from the ground resisting the downward force of a wheel	N
F_{tb}	Tractive force from force versus slip	N
F_w	Net of longitudinal forces acting on a wheel	N
F_{tr}	Longitudinal terrain resistance force	N
F_{ltr}	Lateral terrain resistance force	N
H	Angular momentum	$kg \cdot m^2/s$
H_{nL}	Angular momentum of nth segment left wheel	$kg \cdot m^2/s$
H_{nR}	Angular momentum of nth segment right wheel	$kg \cdot m^2/s$
i	Slip ratio	$percentage$
I	Moment of inertia	$kg \cdot m^2$
$\vec{L}_{1F}$	Vector from segment C.G. to front axle pivot	m
$\vec{L}_{1R}$	Vector from segment C.G. to rear axle pivot	m
$\vec{L}_{2F}$	Vector from segment C.G. to front body pivot	m
$\vec{L}_{2R}$	Vector from segment C.G. to rear body pivot	m
L_a	Distance from axle center to wheel center	m
M	Number of stationary segments during inchworm movement	$unitless$
m	Mass	kg
N	Number of segments in a multi-segment vehicle	$unitless$
n	Segment number from 1 to N	$unitless$
R_w	Wheel radius	m
T_m	Torque from motor	Nm
T_{tr}	Torque from terrain resistance	Nm
T_ω	Torque from drive train friction	Nm
V_n	Velocity of nth segment	m/s
V_v	Velocity of the overall vehicle	m/s
V_w	Longitudinal wheel speed over ground	m/s
$V_\perp$	Lateral wheel speed over ground	m/s
b_w	Wheel width	m
X_t	Terrain coefficient/quality index	$unitless$
X	i position in inertial frame	m
X'	i position in body-fixed frame	m
X_n	i position of nth segment in inertial frame	m
$\ddot{X}$	i acceleration in inertial frame	m/s^2
Y	j position in inertial frame	m

Y'	j position in body-fixed frame	m
Y_n	j position of nth segment in inertial frame	m
$\ddot{Y}$	j acceleration in inertial frame	m/s^2
Z_n	k position of nth segment in inertial frame	m
Z'	k position in body-fixed frame	m
$\ddot{Z}$	k acceleration in inertial frame	m/s^2
α	Angular acceleration	rad/s^2
θ_{axle}	Axle orientation in the inertial frame	rad
θ_n	Segment n orientation in the inertial frame	rad
ϕ_n	Angle between segment n and n+1	rad
τ_{trm}	Terrain resistive moment	$N \cdot m$
ω	Angular speed	rad/s
ω_w	Angular wheel speed	rad/s
ω_{SP}	Angular wheel speed set point; the desired value	rad/s

Abbreviations

$BLDC$	Brushless DC motor; the motor of choice for many UGVs
CSV	Comma-separated value; data storage format similar to a spreadsheet
DE	Discrete Element; numerical modeling technique using discrete elements
DOF	Degree Of Freedom; independent mechanical parameter
FE	Finite Element; numerical modeling technique relying on a mesh of points
$GMDRL$	General Motors Defense Research Laboratories; defense and space division sold to General Dynamics in 2003
$GNSS$	Global Navigation Satellite System; satellite based positioning system
GPR	Ground Penetrating Radar; used to survey subsurface features
GPS	Global Positioning System; US satellite constellation for positioning and navigation, a type of GNSS
$GrIT$	Greenland Inland Traverse; ground traverse from Thule Air Base near Qaanaaq to Summit Station, Greenland
JPL	Jet Propulsion Laboratory; NASA facility focused on robotic spacecraft
LRV	Lunar Roving Vehicle; four wheeled electric rover used on Apollo 15, 16 and 17. Also known as the "moon buggy"
$MARV$	Multi-element Articulated Research Vehicle; built by GMDRL under the guidance of Bekker
NSF	National Science Foundation
ODE	Open Dynamics Engine; default physics engine in Gazebo
PID	Proportional, Integral, Derivative; common type of controller
PSR	Permanently Shadowed Region; specifically near the lunar south pole
ROS	Robot Operating System; open source robotics software architecture
ROV	Remotely Operated Vehicle, teleoperated with little to no autonomy
$SLRV$	Surveyor Lunar Roving Vehicle; built as part of lunar Surveyor program but not flown

SPT	South Pole Traverse; thrice annual tractor convoy from McMurdo Station to Amundsen-Scott South Pole Station
STL	File format for describing 3D surfaces
UGV	Uncrewed Ground Vehicle, land based autonomous robot
$UHMW$	Ultra High Molecular Weight polyethylene; material with very low friction against snow, used in sleds and skis
$URDF$	Unified Robot Description Format; subset of XML used by ROS
WMR	Wheeled Mobile Robot
$XACRO$	XML Macro; a macro language for XML files
XML	Extensible Markup Language; a way to store data structures

Terms and Phrases

Crab Movement	When a vehicle moves in a direction other than where its body is pointed; the way by which crabs walk laterally
Duck Walk	Method of changing body or axle orientation to place wheels on undisturbed soil for better traction
Follower-Segment	Any segment other than the lead segment
Gazebo	3D robotic simulation environment with accurate physics
Grousers	Cleats that provide traction on a wheel or track, typically spanning its full width
High-Centered	When a vehicle's underside is in contact with the ground, and this causes immobilization
In Silico	Testing performed within a simulated environment, within the silicon processor of a computer
Inchworm Movement	Coordinated expansion and contraction to move a body forward; how geometer moth caterpillars moves
Jackknife	When the angle between vehicle segments is forced to its limit due to a mismatch in traction and momentum
LC-130	Ski equipped aircraft in use by the NSF and US Air Force for cargo and personnel transportation in the polar regions
Lugs	Tractive elements on a continuous rubber track. Similar to grousers, but smaller individual segments rather than linear elements spanning the entire track width
Lead-Segment	The front most segment in a multi-segment vehicle
Lebel-sur-Quévillon	Town in Quebec, Canada where mobility testing was performed with *FrostyBoy*
M-to-one	Type of inchworm movement where each moving segment has *M* corresponding segments working to remain stationary
Matlab	Originally a shortening of matrix-laboratory and stylized MATLAB by its creators, it is a combined IDE and interpreted language. It is focused on matrix manipulation and data visualization among much else

Névé	Snow that has metamorphosed into a hard, styrofoam like material
Passive Steering	Method by which an axle with two driven wheels and a central yaw pivot is steered through differential wheel speed
Prismatic	Joint that allows linear sliding about a single axis, like a piston
Revolute	Joint that allows rotation about a single axis, like a hinge
Skeg	Fin or keel added to a sledge for directional stability and to prevent lateral movement
Skid-Steering	Steering done through differential wheel speeds and fixed steering angles. Also called tank-turning
Sledge	Heay duty sled of various designs for carrying cargo over snow
Strandbeest	"Sand beast" walking vehicles built by Dutch artist Theo Jansen. These wind powered kinetic sculptures have excellent mobility on sand
Thrust-Stride-System	Alternative method of wheeled locomotion where wheels are used as feet to push against the ground
Ton	Metric ton; 1000 kg. All units are SI unless explicitly stated
Trafficability	The ability for terrain to support vehicle mobility
Wagon Steering	An axle pivots about its center to provide steering; the angle is often set by a yoke pulled from ahead
WGS-84	Reference ellipsoid used by GPS and many other systems to approximate the imperfect Earth as an ideal mathematical surface

Chapter 1

Introduction and Background

Direct human observation of the natural world was the first form of scientific data collection. However, human senses are limited, and these limits have led to the creation of increasingly sophisticated scientific instruments to measure phenomena that cannot be quantified through direct human observation. Danish astronomer Tycho Brahe built astronomical equipment to allow the human eye to record the movement of stars and planets over the course of decades [1], while more recent advances have allowed for automated measurements without the need for a human observer. Scientific inquiry also extends beyond phenomena that can be measured in a laboratory, and it is often new and challenging locations that present the greatest opportunity for scientific discoveries. The history of exploration is intertwined with scientific data collection and there are countless stories of danger and heroism in the search for knowledge. Explorer Richard Byrd nearly lost his life while operating an Antarctic meteorological station alone during the austral winter of 1934 [2], and while the danger of such activities has decreased, exploring the unknown is never without risks. The polar regions and outer space are two extreme environments that have pushed the development of advanced methods of data collection. Autonomous systems for data collection are one way that humans have decreased the danger and increased capabilities in these extreme environments.

Uncrewed ground vehicles (UGVs) have been used to collect a wide range of scientific data in the polar regions and beyond Earth. Typically wheeled or tracked, UGVs come in a wide range of sizes and masses, from the 23 kg *SnoBot* [3], to NASA's one ton *Perseverance* rover. The four wheeled, 70 kg *Yeti* has used ground penetrating radar (GPR) to peer into ice sheets and survey crevasses [4] [5]. *Yeti* was designed to negotiate *sastrugi*, a pattern in wind-scoured snow that creates linear features similar to frozen ocean waves. The 750 kg *Lunokhod-1* was the first UGV used beyond Earth when it landed on the Moon as part of the Soviet space program [6], and it carried an array of scientific instruments [7]. *Lunokhod-1* was able to explosively disconnect any of its eight wheels in case of damage or immobilization. A major limitation and focus in the design of these vehicles is off-road mobility. The promise of UGVs lies in their ability to make measurements across large distances, and over time periods longer than a crewed expedition could practically handle. Immobilization prevents a vehicle from covering those long distances, and often leads to failures that limit the time available for measurements. Mobility is therefore an overriding consideration that other decisions must work around.

The 2004 Mars Exploration Rovers (MERs) had drastically different mission durations as a result of immobilization, although both far exceeded their planned 90-day missions. *Opportunity* operated until 2018 before losing communication during a dust storm. *Spirit* lost communication in 2010 after an immobilization in 2009 forced it to endure the Martian winter without its solar panels pointed in the optimal direction. If *Spirit* had not become immobilized, it could have provided close to another decade of science data. However, had *Spirit* been designed to make the probability of that immobilization near zero, it would have required compromises to other systems and rendered the extra mission time less valuable.

The next UGV to land on Mars after the MERs was the Mars Science Laboratory (MSL), *Curiosity*, in 2012. *Curiosity*'s ongoing operations must balance the risk of damage to its wheels from sharp rocks with the risk of immobilization in deep sand. Mission planners therefore select routes that straddle sandy and rocky areas, often resulting in routes between areas of interest that are far from direct [8]. *Curiosity* devotes much of its limited mass to scientific instruments, and its mobility system exists in service to those instruments.

My own experience with UGVs is in the polar regions, primarily for performing GPR surveys. Therefore, this thesis will focus on polar applications, but the insights are applicable far beyond, from forests, to deserts, to some moons and other planets. Most UGVs used in the polar regions have been four wheeled, sub-100 kg vehicles. Some have had fully rigid chassis, such as *Cool Robot* [9], while others have had different types of articulation, such as *Yeti* [5] and *FrostyBoy* [10]. *FrostyBoy* will be a major point of comparison throughout this thesis. *Grover* is one notable exception to this formula; it uses twin tracks and weighs over 350 kg. *Grover*'s track system means a low probability of immobilization, but at the cost of high energy consumption, lower speeds, and fewer hours per day of solar powered driving [11].

For all of these vehicles, the mass that can be devoted to their mobility systems must be weighed against other mission needs. A Mars rover could be designed to reduce the probability of immobilization relative to existing UGVs, but would have less payload for sensors - its *raison d'être*. Research has shown that increasing a wheel's width and/or radius results in lower ground pressure, lower sinkage, more traction, and a general decrease in the probability of immobilizations. Such research is summarized in the review paper by Wallace and Rao [12], as well as presented by others [13] [14]. However, these changes require changes to other vehicle systems that further increase mass and negate some of their benefits. For UGVs to be used more widely than they are now, they require mobility improvements that push beyond the simple scaling of wheels, or the lower efficiency of using tracks.

If UGVs were more mobile, able to traverse terrain where there is currently too high a risk of immobilization, then they could be used for applications that are currently beyond their capabilities. For terrain that is currently accessible, decreasing the risk of immobilization would allow travel much farther from crewed bases of operation. Current polar UGV applications are largely limited to survey work over distances of less than one hundred kilometers, and increasing this distance by a factor of ten or more would have profound effects on how UGVs could be used. Current GPR crevasse surveys of the McMurdo Shear Zone (MSZ) could instead cross back and forth over the Ross Ice Shelf, or trace the flow of ice upstream to the glaciers flowing down from the Antarctic Plateau. Instead of automated weather stations being placed by either crewed traverse or aircraft, they could be delivered to remote and inhospitable areas by UGV. UGVs could act as mobile sensor platforms, and a network of such vehicles could spread across an ice sheet to take measurements with higher spatial resolution than any existing ground based measurements. These mobile sensor networks could also help to ground-truth satellite-based measurements, further increasing the benefits from space-based sensors. Ray et al. [9] specifically calls for the creation of long distance sensor networks using UGVs, and the work outlined in this thesis can be viewed as a step toward answering that call.

This thesis presents *SnoWorm*, a vehicle concept for vastly improved UGV mobility on snow. Instead of relying on larger wheels or tracks for improved mobility, it is a composite vehicle made of multiple linked segments. Each segment is similar in scale to the *FrostyBoy* UGV on which it is based, but with active prismatic joints embedded in a tongue assembly between segments. These prismatic joints allow segments to push and pull one another, or to drive in unison.

Figure 1.1 shows the *SnoWorm* rendered within the Gazebo simulation environment, which will be discussed in Chapter 2. The vehicle shown has three bodies (the thicker rectangular prisms), joined by two tongue assemblies (the thinner rectangular prisms). Each body has two axles, and each axle has two wheels (the semi-hollow cylinders).

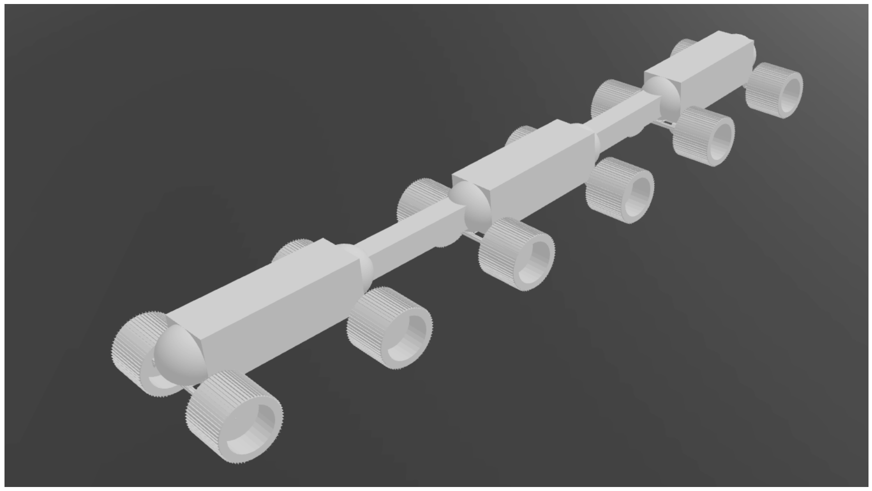

Figure 1.1: The vehicle geometry, called *SnoWorm*, used in the remainder of this thesis.

What follows in Section 1.1 details how existing UGVs have been used in the polar regions. Section 1.1.1 gives a more detailed description of the mobility limitations of those UGVs and Section 1.1.2 gives possible applications of UGVs with improved mobility. Section 1.2 reviews and categorizes a number of vehicles that were considered while working on this thesis. Section 1.3 further describes the *SnoWorm* multi-segment vehicle studied in this thesis. Finally, Section 1.4 summarizes the contributions of this thesis and outlines the remaining chapters.

1.1 Polar UGV Applications

This section gives examples of how UGVs have been used in the polar regions, how their mobility limitations affected those uses, and potential applications that could be addressed if those limitations were removed.

Mankoff et al. [15] used *FrostyBoy* in concert with data from aerial radar surveys [16] and human operated snowmobiles to locate and recover parts of an engine that catastrophically failed during the September 30th, 2017 flight of Air France Flight 66 [17]. Aerial radar surveys combined with ice flow modeling gave a 35,000 m^2 (about 8.5 acres) area believed to contain the engine components. Within this area, *FrostyBoy* performed GPR surveys to locate crevasses and determine a safe route for a human operated snowmobile. The snowmobile towed the heavy *SnowTEM* sensor, which was specifically designed to detect titanium parts buried in snow [18]. The GPR towed by *FrostyBoy* was able to locate the part during a crevasse/route survey, and it was safely extracted after its location was confirmed with the *SnowTEM*. In this case, the original plan was for *FrostyBoy* to tow the *SnowTEM* sensor, but the required drawbar pull was much higher than expected, and *FrostyBoy* was limited to towing the GPR.

UGVs have also been used in the polar regions for purely scientific purposes. Arcone et al. [4] describes a multi-year project using UGVs for GPR surveys in and around the MSZ. The project collected over 1300 kilometers of GPR data and suffered only a handful of immobilizations, all while operating in heavily crevassed terrain that prevents the efficient operation of crewed vehicles. *Yeti*, its twin rover, *Scotty*, and *Cool Robot* were all used in the data collection, and have benefits over the alternative crewed options. Helicopter mounted GPR systems exist, but are more weather dependent, and provide a larger radar "footprint" due to their height above the ground [19]. Crewed vehicles with GPR antennas mounted on long booms are another option [20] but require the human operator to go in harm's way. Arcone et al. [4] used UGVs to collect GPR data to determine the distribution and geometry of crevasses, as well as the thickness and composition of the ice in which they formed. Kaluzienski et al. [21] was another output from this work, and used the GPR measurements of crevasses in conjunction with satellite derived surface velocities to estimate the conditions needed for crevasse formation and propagation.

In addition to *Yeti*'s use in Antarctica, Lever [5] earlier used *Yeti* as a route finding tool for the Greenland Inland Traverse (GrIT). Sent out ahead of the tractor convoy, *Yeti* once again towed a GPR to survey hidden crevasses and help find a safe route through difficult terrain. GrIT helps to bring liquid fuel, supplies, and equipment from Thule Air Base on Greenland's coast to Summit Station at the top of the Greenland Ice Sheet. The alternative to moving cargo with GrIT is the use of LC-130 ski equipped cargo planes, which must